

JANHAVI GAIKWAD

Ithaca, NY(Open to Relocate)|100K Not Required | janhavigaikwad17@gmail.com | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Cornell University

Ithaca, NY | May 2026

Master of Engineering in Systems Engineering

Relevant Coursework: Model-Based Systems Engineering, Systems Architecture, Feedback Control Systems, Digital Twin, Data Science & AI for Systems Engineering, System Analysis & Behavior Optimization, Project Management

Priyadarshini College of Engineering

Nagpur, India | November 2020

Bachelor of Engineering in Aeronautical Engineering

Relevant Coursework: Computational Fluid Dynamics, Aerodynamics, Aircraft Propulsion, Structures, C

WORK EXPERIENCE

Honeywell Technology Solutions

Bengaluru, India | Jan 2024 – Jul 2025

Mechanical Design Engineer

- Delivered 300+ hours of engineering productivity gains by designing and validating in-house OD gas-turbine cycle performance algorithms in C++ (OOP), developing compressible flow thermodynamics and gas dynamics correlations, gas property corrections, trap efficiency, and scalar logic across fan, compressor, turbine, and combustor modules.
- Reduced simulation setup time by ~5 minutes per run by developing UI-based access and 3D aerodynamic performance map visualization, improving engine modeling efficiency and usability for propulsion system analysis.
- Maintained feature accuracy within 0.05% tolerance by validating turbojet and turbofan engine models in NexSys against FAST legacy models, ensuring reliable propulsion performance assessment across design configurations.
- Achieved >95% say-do ratio in a global agile engineering environment through disciplined planning, ensuring consistent and reliable delivery of high-fidelity propulsion modeling features.
- Enhanced team-wide efficiency by authoring monthly Kaizen documentation capturing micro-optimizations, simulation shortcuts, and engine modeling insights across gas-turbine system workflows.
- Supported cross-functional engineering studies by modifying engine performance models across varied propulsion configurations, contributing analytical insight to system-level efficiency trade analyses.
- Developed a proof-of-concept ML system for electric UAM route optimization, enabling early-stage propulsion and mission-efficiency studies projected to reduce energy consumption by ~10%.
- Developed a proof-of-concept ML model for gas turbine compressor surge and stall prediction, supporting operability risk assessment for next-generation propulsion system design.

Entuple Technologies Ltd.

Pune, India | Aug 2022 – Jan 2024

CFD Application Engineer (ANSYS)

- Enabled ANSYS Fluent adoption across aerospace, automotive, and research clients by providing expert CFD simulation support, troubleshooting, and aerodynamic model validation guidance for 50+ customers.
- Improved simulation proficiency by ~40% for 500+ engineers through delivery of 20+ CFD technical training programs in ANSYS Fluent, achieving a 4.8/5 satisfaction score and reducing aerodynamic modeling errors by an estimated 25%.
- Accelerated customer adoption by developing 10+ proof-of-concept CFD models spanning multiphase flows, pollutant dispersion, chemical reaction modeling, and moving/dynamic mesh simulations for aerospace and automotive applications.
- Resolved complex CFD challenges for strategic clients across aerodynamic and thermal domains — providing simulation best practices, mesh setup guidance, and validation support, resulting in ~20% faster model turnaround.
- Developed Fluid-Structure Interaction (FSI) simulation models in ANSYS coupling fluid pressure loading with structural response, validating component design integrity under realistic aerodynamic loading conditions.
- Delivered a keynote at ICPHD'23 (IIT Guwahati) showcasing ANSYS simulation tools for green energy and sustainability engineering applications, reaching 100+ conference attendees.

Hindustan Aeronautics Ltd.

Nashik, India | Aug 2021 – Aug 2022

Apprentice Trainee

- Achieved <1% error in aerodynamic lift and drag predictions under operational flight conditions by implementing a MATLAB-based system identification framework applied to flight data recorder (FDR) data.
- Optimized CFD meshing strategy for complex aircraft geometries by conducting a comparative study across multiple tools and techniques, evaluating aerodynamic solution accuracy and computation time to streamline the analysis workflow.
- Performed CFD simulations of aircraft and airfoil geometries using ANSYS Fluent and CFD++, enabling comparative aerodynamic analysis, design validation, and flight performance assessment.

- Conducted an IPR awareness lecture at HAL, promoting innovation culture and research compliance within the aeronautical engineering organization.

Indian Institute of Technology Madras

Chennai, India | May 2019 – Jul 2019

Summer Research Fellow

- Accelerated CFD preprocessing efficiency by developing an optimized FORTRAN algorithm, reducing computation time for 600,000-element aerodynamic meshes from 50–60 minutes to 1–2 minutes (~97% reduction).

ACADEMIC PROJECTS

EKF-Based Gas Turbine Engine Health Monitoring

- Developed a 7-state reduced-order nonlinear gas turbine thermodynamic model (compressor, combustor, turbine, shaft dynamics) integrated with a discrete-time EKF to estimate compressor outlet temperature, combustor temperature, turbine outlet temperature, shaft speed, and component efficiencies (η_c , η_b , η_t) as health indicators from 4 noisy cruise measurements.
- Derived analytical Jacobians (F_k , H_k) for EKF prediction/update cycles and modeled efficiency degradation as slow random-walk states, enabling gradual wear tracking without direct sensing; validated estimator consistency with $\pm 2\sigma$ covariance bounds and confirmed full local observability across all 7 states.
- Demonstrated reliable state estimation under noisy cruise conditions, including inference of unmeasured turbine outlet temperature T_4 purely through model dynamics, with estimation errors settling within $\pm 2\sigma$ bounds after brief initial transients.

Systems Architecture for Next-Generation Aviation Aircraft

- Led end-to-end system architecture development for a hybrid-electric general aviation aircraft by translating stakeholder needs into 20 traceable program goals and measurable system requirements.
- Developed SysML-based functional, operational, and physical architectures, including state machines, use cases, BDD/IBD models, allocated architectures, and a 19-component DSM, to define subsystem interactions, LRU partitioning, and integration constraints prior to detailed design.
- Designed a constraint-driven architecture exploration framework that enumerated and evaluated 184,000+ feasible aircraft configurations using full-factorial enumeration across the constrained design space.
- Applied multi-objective tradespace exploration, Pareto optimization, and genetic algorithms to identify high-value architecture families and enable data-driven aircraft design selection.

Frequency-Domain SISO Turbine Control

- Designed a frequency-domain SISO controller in MATLAB for an academic turbine system, minimizing inlet flow disturbances on spool speed while maintaining reference tracking and ensuring closed-loop stability across the operating envelope.

Inverted Pendulum Control Design

- Designed, implemented, and experimentally validated a feedback controller for an inverted pendulum in both time and frequency domains, achieving required stability margins, disturbance rejection, and settling time performance.

3D Urban Air Mobility (UAM) Path Planning

- Designed and implemented 3D UAM path planning frameworks over Hong Kong city airspace with no-fly zone constraints, evaluating Visibility Graph + Dijkstra and buffered 3D A* routing architectures.
- Delivered a structured trade study comparing path optimality, computational cost, and scalability to support architecture selection for autonomous urban-scale aerial routing systems.

Airport Preboarding System Analysis & Optimization

- Developed a Python-based discrete-event simulation of airport preboarding operations, modeling passenger flow across check-in, transit, security, and boarding stages.
- Performed Monte Carlo simulation and statistical analysis to evaluate queue behavior, throughput, and system stability; conducted staffing trade study identifying resource configurations that reduced operational cost while maintaining target throughput.

MBSE Project: Solar-Aware Route Optimizer

- Delivered a user-centered system architecture by translating 6 prioritized customer needs into measurable requirements using Customer Affinity Analysis and Customer Value Proposition (CVP) frameworks.
- Reduced requirements ambiguity by formalizing system behavior via SysML use case, activity, sequence, and state diagrams; improved design decision efficiency by 20% through AHP, QFD, FMEA, and fault tree analysis.
- Ensured verification readiness and lifecycle traceability by mapping derived requirements to test strategies via SysML requirements diagrams and a Verification Cross-Reference Matrix (VCRM).

Solar-Aware Route Optimizer – Digital Twin

- Built a Unity-based digital twin of Manhattan by importing six city sections from government GIS datasets and integrating building and road data into a unified interactive model for city-scale solar-aware route planning.
- Developed a C# algorithm within Unity to construct a city-wide road network, computing road lengths and endpoints to evaluate solar shadow parameters; implementing optimized route-finding using Visibility Graph, Dijkstra, and EJOB optimization.

Smart Charge App – EV Charging Optimizer

- Developed a real-time EV charging decision-support system converting live grid carbon intensity data into low-emission charging schedules, integrating data APIs, LLM-based reasoning, and calendar systems into an end-to-end pipeline.

News Intelligence Dashboard

- Built a real-time news intelligence dashboard with AI-generated summaries, sentiment inference, and a multi-agent Signal Studio pipeline for cross-section narrative synthesis; deployed end-to-end on Posit Connect.

Supply Chain Dashboard – SupplyMindAI

- Built a live supply chain operations dashboard with an agentic AI advisory layer supporting natural-language queries, simulation-backed decision support for delay/capacity/routing trade studies, and real-time shipment visibility.

Project Management Plan – Vertical Farming Enterprise

- Created a comprehensive project management plan including resource planning, risk assessment, FMEA, and Gantt charts to enable streamlined execution of a vertical farming enterprise program.

CERTIFICATIONS

ANSYS Fluent – Computational Fluid Dynamics | Ansys Issued 2023

ANSYS Workbench – Product Design & FEA (Structural,Thermal Analysis, FSI Analysis) Issued 2020

TECHNICAL SKILLS

Propulsion & Simulation: OD gas-turbine cycle performance modeling · Compressible flow & gas dynamics · Aircraft propulsion & operability · System identification · CFD (ANSYS Fluent, CFD++, STAR-CCM+, FORTRAN) · FEA – Structural & Thermal Analysis (ANSYS Workbench) · Fluid-Structure Interaction (FSI) · Multiphase flow · Moving/dynamic mesh · Aerodynamic analysis · Compressor/turbine/combustor modeling

Systems Engineering: Systems architecture · Tradespace & Pareto analysis · Requirements engineering · Verification & validation · MBSE (SysML, QFD, FMEA, FFBD, IDEFO, CVP, VCRM) · Discrete-event simulation · Project management (WBS, EVM, risk)

Controls & Estimation: Feedback control (time & frequency domain) · State-space design · Extended Kalman filtering · State estimation · Model-based design

Programming & Data: Python · MATLAB · C++ (OOP) · C · FORTRAN

AI & Applications: Agentic AI · LLM integration · Tool calling · Multi-agent workflows · REST APIs · Machine learning (POC) · Shiny for Python

Tools & Platforms: Cameo · ANSYS Fluent · ANSYS Workbench · ANSYS Meshing · CFD++ · STAR-CCM+ · NexSys · FAST · Simulink · CATIA V5 · STK · Visual Studio · ProjectLibre · Posit Connect · GitHub